WIRELEES NETWORKING

1.List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

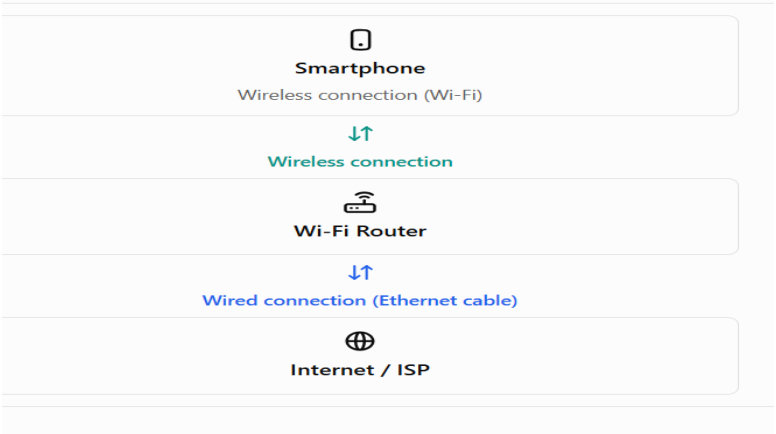
ANS:

Wi-Fi Standard	Frequency Band	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Speed up to 600 Mbps.
802.11ac (Wi-Fi 5)	5 GHz	Faster speed, up to about 3.5 Gbps.
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz	Speed up to 9.6 Gbps and better performance with many devices.

2.Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

ANS:

Home Wi-Fi Network Diagram



Smartphone

Wireless connection (Wi-Fi)

Wireless connection

Wi-Fi Router

Wired connection (Ethernet cable)

Internet / ISP

Explanation:

- Wireless part: The smartphone connects to the Wi-Fi router using radio waves.
- Wired part: The router connects to the modem/ISP using a cable, usually Ethernet between the router and modem.
- The internet connection allows the smartphone to access websites and online apps.

**3.Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Hint: Consider both signal strength and security when choosing.**

ANS:

Wi-Fi Network Scanning Report

Aim: To scan available Wi-Fi networks and compare their signal strength and security type.

Observation Table:

No.	SSID (Wi-Fi Name)	Signal Strength	Security Type
1	Home_WiFi	Strong (90%)	WPA2
2	Neighbor_WiFi	Medium (60%)	WPA2
3	Public_WiFi	Weak (30%)	Open (No Password)

\Wi-Fi Network I Would Choose:

I would choose Home_WiFi because it has a strong signal and WPA2 security. It provides a stable connection and protects the network with encryption. If WPA3 is available with a similarly strong signal, I would prefer WPA3.

4.Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

ANS:

SSID Hiding in Wi-Fi Networks

SSID hiding means disabling the broadcast of a Wi-Fi network's name so it does not normally appear in the list of available Wi-Fi networks.

- Advantage: It makes the Wi-Fi name less visible to casual users.

- Disadvantage: It does not provide strong security because the network name can still be discovered using special tools, and users must enter the SSID manually to connect.